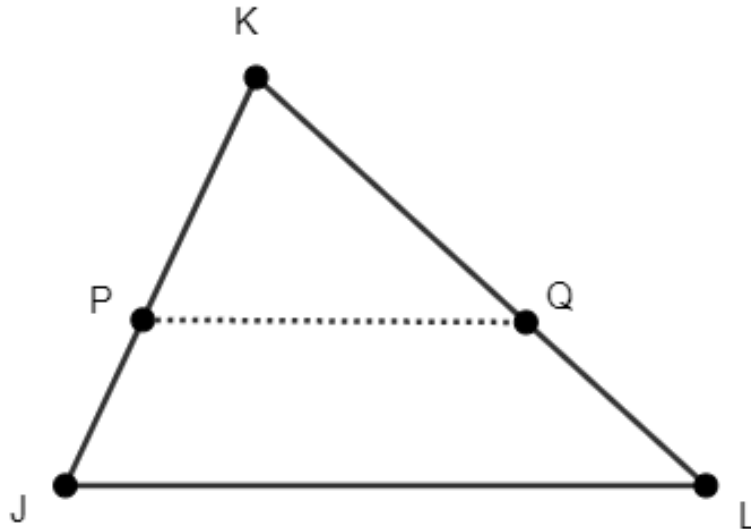


$\triangle KJL$ is shown, where $m\angle KJL = 65^\circ$, $m\angle KLJ = 42^\circ$, and $m\angle JKL = 73^\circ$. Point P is on $\overline{KJ}$ and Point Q is on $\overline{KL}$. $\overline{PQ}$ is drawn as a dotted line.



1. Use a ruler and a protractor to complete the table.

Measurements of $\triangle JKL$		Measurements of $\triangle PKQ$	
$JK = 6 \text{ cm}$	$m\angle KJL = 65^\circ$	$PK = 3.5 \text{ cm}$	$m\angle KPQ = 65^\circ$
$KL = 8 \text{ cm}$	$m\angle JKL = 73^\circ$	$KQ = 4.6 \text{ cm}$	$m\angle PKQ = 73^\circ$
$JL = 8.5 \text{ cm}$	$m\angle KLJ = 42^\circ$	$PQ = 5 \text{ cm}$	$m\angle KQP = 42^\circ$

2. The corresponding angles of $\triangle KJL$ and $\triangle KPQ$ are $\angle K$ and $\angle K$, $\angle J$ and $\angle P$, $\angle L$ and $\angle Q$.

3. Complete the table.

Sides	Ratio
Determine the ratio of JK to PK .	$\frac{6}{3.5} = 1.714$
Determine the ratio of KL to KQ .	$\frac{8}{4.6} = 1.739$
Determine the ratio of JL to PQ .	$\frac{8.5}{5} = 1.700$

4. The corresponding sides of $\triangle KJL$ and $\triangle KPQ$ are $\overline{JK}$ and $\overline{PK}$, $\overline{KL}$ and $\overline{KQ}$, $\overline{JL}$ and $\overline{PQ}$.

5. $\triangle KJL$ is similar to $\triangle KPQ$.
6. Describe a sequence of transformations that can be used to map $\triangle KJL$ onto $\triangle KPQ$.
 $\frac{1}{1.7} \approx 0.59$
 Dilation with K as the center of dilation and a scale factor of 0.59.

Complete each relationship for $\triangle KJL$ and $\triangle KPQ$.

7. $\frac{PK}{JK} = \frac{PQ}{JL}$

8. $\frac{JK}{PK} = \frac{LJ}{QP}$

9. $\frac{KQ}{KL} = \frac{KP}{KJ}$

10. Observing that $\frac{JL}{PQ} \neq \frac{LQ}{QK}$, your classmate Moriah claims that $\triangle KJL$ and $\triangle KPQ$ are not similar. If her logic is faulty, explain why.
 $\overline{LQ}$ is a portion of one side of $\triangle KJL$ and not a corresponding side to $\overline{QK}$, so her logic is faulty.